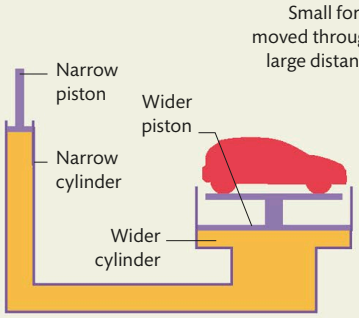
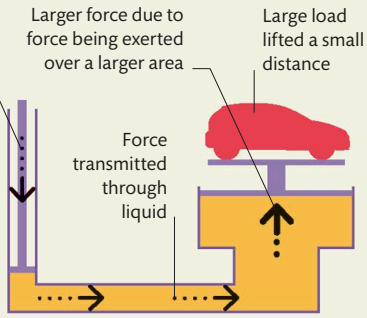
Hydraulics

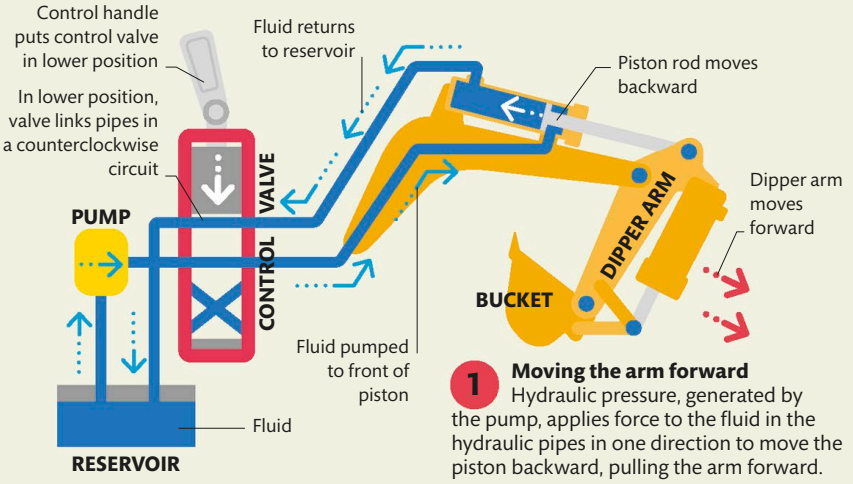
Liquids cannot be compressed (unlike gases), which means that any force or pressure applied to a liquid is transferred through it. In a basic hydraulic system, when pressure is applied to one end of a liquid inside a closed pipe or cylinder, the force is passed all the way to the other end. A small force can be multiplied by changing the width of one piston and cylinder column relative to the other.



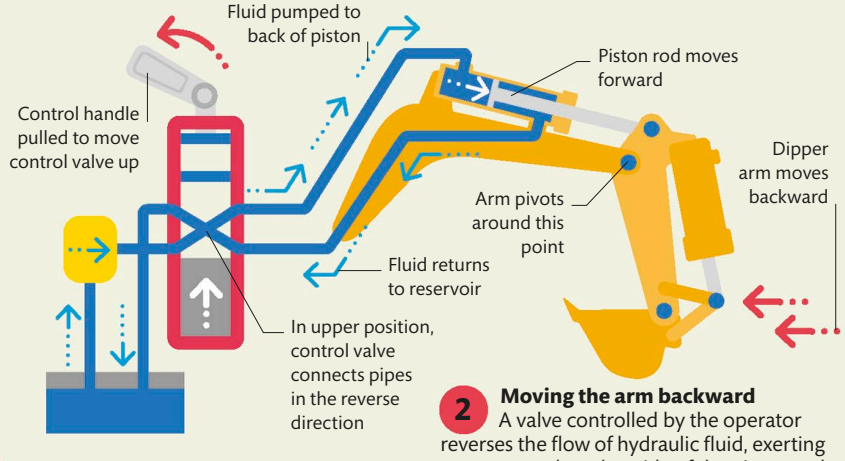
1 Multiplying a force
A force applied by a piston in a narrow cylinder is multiplied into a larger force by a wider piston at the other end, although the liquid's pressure remains the same.



2 Double the force, half the distance
If the larger piston has twice the area of the small piston, the force exerted will double. The cost is that this greater force operates through half of the distance.



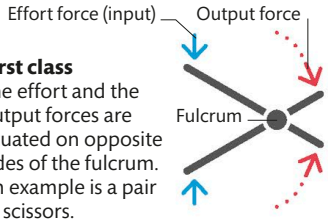
1 Moving the arm forward
Hydraulic pressure, generated by the pump, applies force to the fluid in the hydraulic pipes in one direction to move the piston backward, pulling the arm forward.



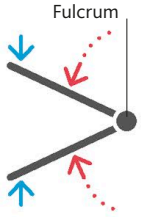
2 Moving the arm backward
A valve controlled by the operator reverses the flow of hydraulic fluid, exerting pressure on the other side of the piston and pushing the arm in the opposite direction.

LEVERS

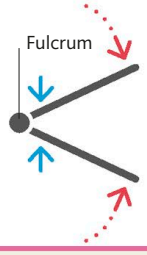
There are three classes of levers, defined by where the effort and output forces are located relative to the fulcrum. They can be used to increase either power or movement in different directions.



First class
The effort and the output forces are situated on opposite sides of the fulcrum. An example is a pair of scissors.



Second class
The output force is located between the fulcrum and the effort force. An example is a pair of nutcrackers.



Third class
The effort force is applied between the fulcrum and the output force. An example is a pair of tongs or tweezers.